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China Offshore Intelligent Oilfield Production Monitoring System: Design and Technical Path Forward for Implementation

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Abstract

The advent of the Fourth Industrial Revolution has precipitated a digital metamorphosis in oil companies, presenting a singular path towards cost reduction and heightened efficiency. As a pivotal direction, the intelligent oilfield represents an intricate, integrative information system. Its capacity to orchestrate petroleum exploration, production decisions, and other relevant sectors through data technology facilitates the standardization and automation of oilfield exploitation. In pursuit of this goal, we propose a novel digital technology solution specifically designed for offshore oilfields. Harnessing state-of-the-art computer technologies such as big data, cloud computing, edge computing, optical fiber communication, and IoT, our aim is to establish an inclusive data integration platform and deliver unified data services to enhance the efficiency of information utilization.

Within the landscape of the digital oilfield, offshore production monitoring and optimization emerges as a promising sector where data processing and advanced analytics can be employed to boost revenue, curtail costs, and mitigate risks. Constructing a data mining and analytics engine to utilize data from wells, equipment, facilities, and field personnel, however, poses a formidable challenge. The workflow of data acquisition from offshore production platform groups—comprising main production platforms, sub-production platforms, transport vessels, and FPSOs—along with the transmission, processing, and timely analysis of data, are major factors in formulating a data service solution. Consequently, an integrative system encapsulating all necessary tools can significantly expedite addressing these challenges in a timely manner.

This paper provides a comprehensive technical elucidation of the currently devised intelligent oilfield production monitoring system. The data service system encompasses four subsystems: (I) Production management, encompassing remote operation monitoring, production dynamic tracking, reservoir dynamic management, and well pattern injection and production deployment management; (II) Equipment and facilities management, which includes equipment health management, submarine pipeline operation and maintenance management, submarine cable management, and inventory management; (III) Scheduling management, integrating personnel dynamic dispatching, ship dynamic dispatching, and meteorological and hydrological monitoring; and (IV) Safety management, ensuring operational safety and personnel status and location management.

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These subsystems are seamlessly integrated and engage in intercommunication, functioning as microservices, cloud-based container services, or edge computing services. They provide a comprehensive solution for real-time offshore production optimization and monitoring. The edge-cloud collaborative system, established on this blueprint, has undergone successful testing and application in a Bohai oilfield.

This paper will be beneficial for oil and gas organizations contemplating the implementation of an offshore intelligent oilfield as part of their digital transformation strategy. Moreover, it will offer valuable insights to any parties interested in acquiring a deeper understanding of intelligent oilfields.

Keywords: Intelligent Oilfield, Production Monitoring System, Production Management, Equipment and Facilities Management, Safety Management

Background

Ever since China's transitioned from being a net oil and gas exporter to a net importer, the country's reliance on external sources for oil and gas has steadily increased, with dependencies exceeding 70% for oil and 40% for gas. This rise has markedly increased pressure on the oil and gas production industry to ensure secure energy supply. Concurrently, the quality of discovered oil and gas resources is continuously declining, and the growth of newly discovered reserves struggles to replace the depletion of used reserves. This predicament represents a persistent issue that domestic oil and gas field production enterprises must confront. At this stage, the global energy industry is transitioning towards intelligent operations. For the marine engineering industry, particularly unmanned, digital, and intelligent offshore oil and gas production, this transformation is an inevitable future trend. This shift also offers a valuable opportunity for oil and gas field enterprises to undertake industrial upgrades and transformation.

With the mature application of digital and informational technologies in the field of oil and gas production, and the continuous increase in digital infrastructure investment by oil and gas field production enterprises, the establishment of digital capabilities has become a vital step for these enterprises to enhance their management and production capacities (Ref.1). Enhancing the digital and informational levels and promoting the deep integration of modern information industries with traditional petroleum industries, has become an essential choice and primary path for innovative development (see Figure 1).

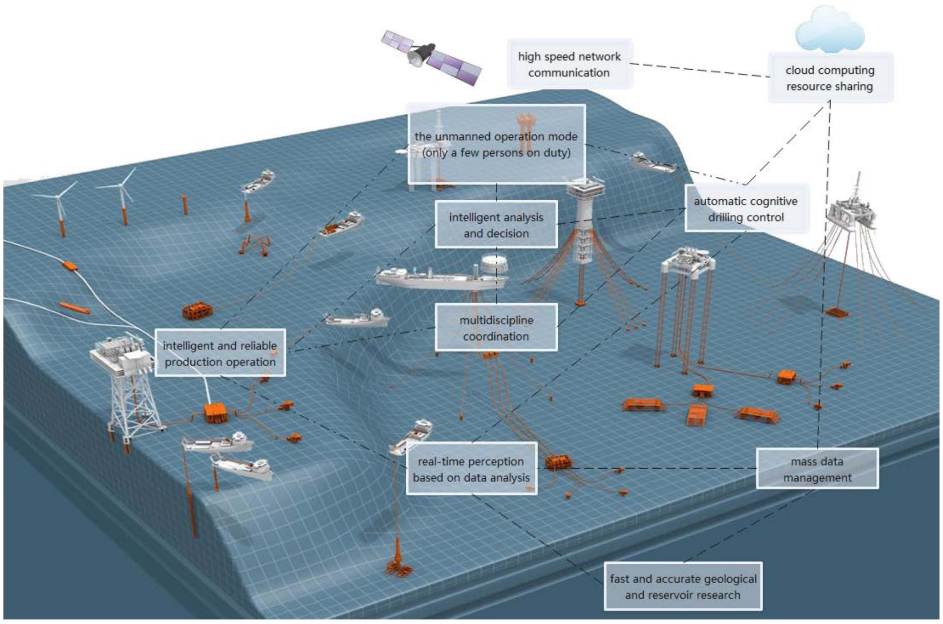

Figure 1—Overview of digital construction in offshore oil and gas production

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The digital construction and intelligent application research and development in the offshore oil exploration and production industry are reflected in the following four aspects (Ref.2):

1. The adoption of Industrial IoT technology seamlessly connects production information systems with oilfield production equipment and facilities, enabling real-time perception of the production process, automatic data collection, real-time analysis, automatic monitoring, and even automatic control. This process has facilitated the transition of traditional production methods, significantly improved production efficiency, and effectively reduced safety production risks.
2. The application of platformization and cloud platform services has emerged as a new model for informational construction. By building platforms such as infrastructure cloud, exploration and production management system, and a geological engineering integrated system, the unified deployment and shared application of various software and hardware resources are realized.
3. The construction of enterprise-level data lake projects and the application of big data analysis techniques have become new methods for companies to enhance their decision-making capabilities. The data lake integrates different types of data, making many previously unanalyzable data available for production without predefined models, thus providing technical support for shared analysis of multi-source heterogeneous data. By big data analysis techniques, users can mine data resources through association analysis and intelligent methods, intuit and analyze data content, and achieve precise correlation descriptions of underground oil reservoirs and surface equipment facilities, enhancing the understanding of oilfield production status and the decision-making support capacity of the company.
4. Mobile internet has become a new approach to comprehensively enhance the capability of informational support services. By utilizing mobile internet technology, the extension of informational support services in time and spatial is realized, providing mobile application support for oil and gas field production operations and full-business management at any time and any place.

As a major offshore oil and gas production and sales enterprise in China, CNOOC initiated the planning study of the "Digital Oilfield" in 2014, completed the construction of a private cloud platform in 2018, and officially released the "digital transformation top-level design outline" in 2020. This document delineates the next phase of digital transformation vision: based on centralized and unified data asset management, the group aims to establish perceptive insights, intelligent control, collaborative sharing, and interconnected innovative digital capabilities. Building a digital ecosystem that equipment and facilities integration will create an intelligent offshore oil company. This transformation will enhance the company management control capabilities, promote cost reduction and efficiency, mitigate significant risks in production and business management, assist in high-quality development, and achieve a leap from traditional management models to modern, digital, and intelligent models.

Furthermore, through the deep integration of digital technologies like cloud computing, the IoT, big data, artificial intelligence, and core exploration and production businesses, the transformation promotes changes in production methods and optimizes management processes. Based on four typical scenarios, namely unmanned operation, visualized oil reservoirs, collaborative operations, and scientific decision-making, the intelligent oilfield application system construction is carried out (Ref.3).

Unmanned operations, primarily targeting production sites, apply digital technologies like IoT, big data, and artificial intelligence to realize remote monitoring and predictive maintenance. The composite services of these technologies continually enhance the automation and intelligence levels of major production sites like offshore platforms and land terminals, reduce the number of personnel at production sites, and promote changes in production operation methods.

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Visualized oil reservoirs technique mainly serves collaborative research by establishing a multi-disciplinary visual oil reservoir research environment. This facilitates the automation of the research process, improves exploration research efficiency, and shortens the exploration discovery cycle.

Collaborative operations primarily focus on the construction of production operation command centers, gradually realizing integrated collaborative operations across specialties, regions, and levels, improving management efficiency, and reducing operation costs.

Scientific decision-making primarily uses intelligent analysis technology to integrate cross-specialty data and multi-dimensional information, providing a supportive environment for key strategic decisions like mineral rights acquisition, oil and gas exploration, oil and gas exploitation and production, and oilfield abandonment.

Based on these plan objectives, the design and investment construction of the offshore intelligent oilfield system is carried out. A step-by-step construction of unmanned or minimally manned platforms based on IoT perception is planned, aiming to build the "smart brain" of offshore oil and gas production and create a coordinated "interconnected oilfield" for sea and land (Ref.4). The overall goal is to fully build offshore intelligent oilfields: based on real-time automatic data collection, supporting the intelligence of core businesses such as exploration, drilling, completion, production and engineering. Through management transformation and process optimization, a comprehensive perception, automatic control, optimized decision-making, intelligent prediction intelligent oilfield system is established. To meet the networking, information exchange, business collaboration of the information system, and to achieve the goal of intelligent oilfield construction, considering the integrated application and development trend of the new generation of information technology, a data-driven, project-oriented, object-oriented full life-cycle management system is established around the full life cycle of oil and gas fields, from resource exploration, resource evaluation, oil and gas discovery, oil and gas reservoir evaluation, development and production, operation management to abandonment. This constructs a full process management from scheme design, project implementation to production operation (see Figure 2).

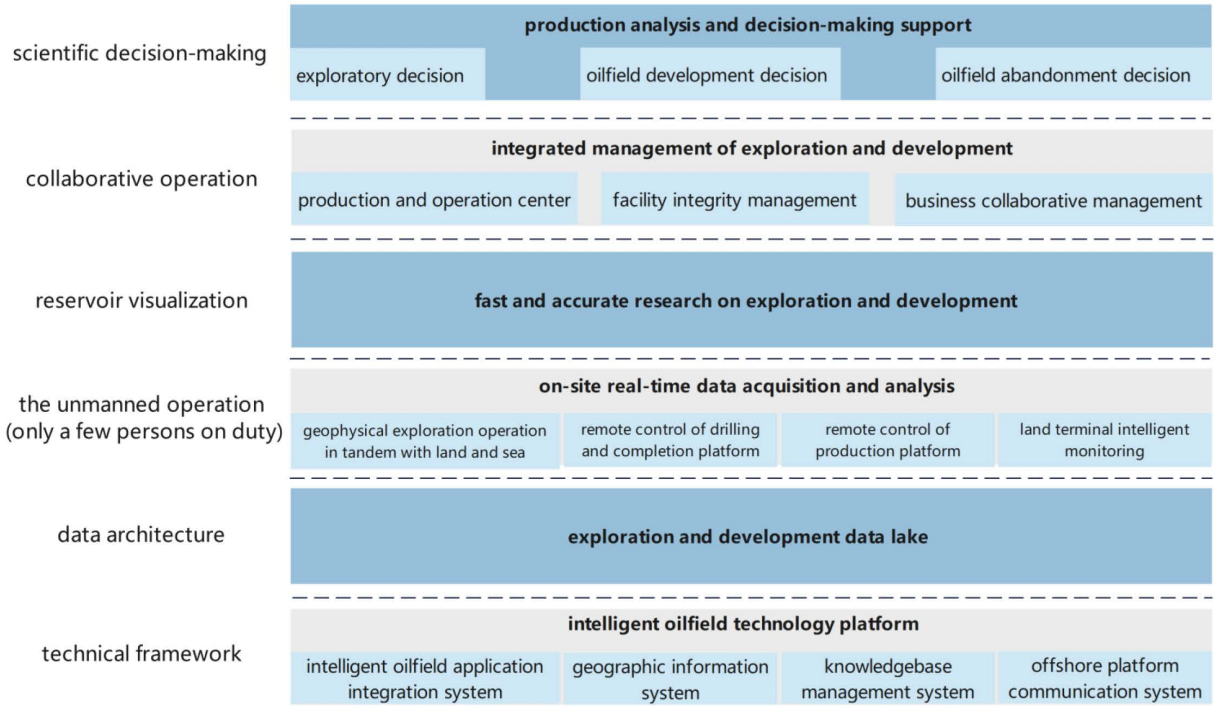

Figure 2—General framework of intelligent oilfield construction

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Challenges and problems

Our company began the design and development of an offshore intelligent oilfield production monitoring system in 2021. As a crucial part of intelligent offshore oilfield construction, this system utilizes industrial internet technology and data analysis to supervise and enhance offshore oil and gas production performance. It can provide real-time status information about wells, pipelines, equipment, and processes. Additionally, it enables remote control and coordination of operations and calculates inferred data not directly measured by physical instruments, such as virtual meters. This production monitoring system assists operators and engineers in optimizing production, reducing costs, enhancing safety and reliability, and minimizing environmental impacts.

However, with the expansion of offshore oil and gas field exploration and production, its model has gradually shifted from being based on a single oil and gas field (a single platform or multiple platforms) to multiple oil field regions. This shift has created a connected network of oil, gas, water, and electricity supply across oil and gas fields. Yet, the production process control systems (PCS, DCS, SCADA, etc.) are still constructed and operated based on the model of a single platform or a master platform with a number of satellite platforms. This substantially increases the complexity of the digital system design for oil and gas production and transportation management. To further reduce the construction cost and development difficulty of the offshore intelligent oilfield production monitoring system, we adopted an implementation strategy based on "steady advancement, pilot first, top-level design, comprehensive planning." We selected the X oil and gas field production operation group in Bohai bay as the pilot construction unit for the production monitoring digital system.

In terms of production operation equipment and facilities, this operation group includes eight WHP, four central platforms, an EPP, and one FPSO. It has a relatively independent production and supply pipeline system. Regarding the fundamental network environment, after network upgrade and expansion, a network environment supported by fiber optics, microwaves, and satellites was built, providing ample bandwidth for future data transmission, sharing, and analysis. Concerning data support, the offshore platform central control systems are highly automated, and many central control subsystems support the universal OPC protocol. PLC/RTU devices provide the standard Modbus protocol, meeting the requirements for real-time data processing for oil and gas production and equipment services. Moreover, the company's self-developed systems based on the C/S structure can satisfy application requirements for oil and gas production and equipment planning, and dynamic data analysis.

Despite the above advantages and a good foundation for the construction of an offshore intelligent oilfield production monitoring system in the X oil and gas field production operation group, there are four main challenges and problems:

1. Field production data has not yet been integrated into a data lake, affecting real-time access and the rapid grasp of front-line production conditions by operators. Currently, offshore production and centralized control real-time data are only preserved separately on various platforms and business departments, lacking data sharing. Information such as production data, equipment information, and vessel demands are not collected, summarized, and centralized in a timely and comprehensive manner, making it challenging to visualize and conveniently view real-time operational dynamics.
2. Offshore platform oil and gas production management lacks effective automated processes and business model support, resulting in low operational efficiency. Due to the lack of an integrated information system for oil and gas production, effective control mechanisms are only reflected in paper documents, Telephone and email communication. Production management requires information about well production dynamics, real-time equipment operation status, marine pipe operational conditions, and equipment facilities archives, among others. These are hard to collect due to their number and real-time requirements.

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3. Daily operations lack support from a unified information system platform. Due to the lack of an information system for support, large amounts of operation data, such as oil and gas production volume and equipment operating parameters, need to be manually recorded and summarized, making it hard to ensure data timeliness and quality.
4. The control systems on offshore platforms, such as central control systems and PLC systems, generally operate in a closed internal network environment. Production data, video surveillance data, and power data from offshore platforms are not effectively and persistently stored. The data required for the production management system is mainly entered manually. There's a lack of an intelligent application system that provides multi-source data services, making it difficult to meet the actual needs of production management in terms of data analysis breadth and depth.

System design

System construction objectives

The offshore intelligent oilfield production monitoring system mentioned in this article targets the actual pain points and difficulties of offshore oil and gas production in the target oil and gas block, focusing on key breakthroughs. Based on the company's unified data management standards and information system construction requirements, the construction approach is centered on ‘cloudification + platformization + cloud-edge collaboration + agile development and delivery.’ The system development employs the DevOps model, employing a cloud-edge architecture construction mode and a ‘data + platform + modular application’ framework.

In combination with actual business needs, the aim is to aid precise decision-making, enhance sea-land collaboration and production operation efficiency, reduce production costs, and realize the construction of the industrial internet of offshore oil and gas production. The system primarily achieves the following objectives:

1. One platform: A basic operation and maintenance platform is established that suits a unified environment of cloud-edge collaborative services. It realizes the centralized, unified, and precise management of information resources, ensures the reasonable allocation of resources, provides a standardized, secure, and open public service environment for upper-level business applications, and prevents repetitive software function development to improve IT resource usage efficiency.
2. Intelligent reservoirs: Production monitoring models, dynamic evaluation models, and production policy optimization models are created based on the fusion of dynamic and static data of oilfields. This replaces the traditional manual data collation and various commercial software for data processing and computation, realizing platform operation for dynamic analysis, indicator evaluation, potential tapping of oilfields, and adjustment optimization.
3. Intelligent injection and production: Based on data fusion of injection and production, work condition monitoring models, evaluation models, and measure optimization models are constructed. So as to improve the service abilities of the work scenarios such as dynamic monitoring of oil or water well, abnormal monitoring and analysis of oil or water well working conditions, and formulation of oil and water treatment measures.
4. Intelligent production operation: By integrating production data with process flow data, various models are established, including production monitoring, operation alarm, platform online process flow, and wellhead virtual metering models.
5. Predictive maintenance of equipment and facilities: Utilizing a combination of equipment and facilities archive data, operation data, and production data, we construct fault diagnosis models for equipment and facilities, as well as equipment health analysis models.

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6. Intelligent safety and environmental protection: Risk alerting models and evaluation models are created based on the fusion of equipment and facilities archive data, detection data, operation data (including video data), and production data.

System architecture design

The system is built upon the investment in information infrastructure of offshore oil and gas fields. It revolves around the core assets of oil and gas fields, such as oil reservoirs, oil-water wells, pipelines, and production facilities. Leveraging deep information technology to assist asset management and benefit optimization, the system associates reservoir data, production data, and safety monitoring data, thereby supporting dynamic analysis of reservoir research and production management. It enables two-dimensional and three-dimensional comprehensive displays of reservoirs, well networks, production processes, and facilities, thereby building capabilities for real-time perception, integrated coordination, monitoring and prediction, and analysis and optimization, thus helping in high-quality operation, and maximization of offshore oilfield asset value.

The design of the system's information architecture focuses on promoting intensive construction, information sharing, and business collaboration. It comprehensively considers the informatization development of offshore oil and gas field production, including six aspects of architectural design: overall architecture, application architecture, technical architecture, data architecture, integration architecture, and network security architecture.

Overall architecture. The offshore intelligent oilfield production monitoring system comprehensively evaluates the unique aspects of both offshore and onshore domains where the edge end implements data collection and real-time analysis of the monitoring system of offshore platforms, and the cloud end mainly supports various intelligent business applications. Its overall architecture is shown in Figure 3. To avoid problems such as unshareable software code, difficult reuse of functional modules, difficult docking of data resources, and complicated maintenance and upgrade difficulties caused by inconsistencies in software architecture and development specifications, the concept of "systematic design, modular development, standardized construction, and integrated assembly" is adopted. A unified technical architecture focusing on data-driven foundation, and business applications has been established, which puts forward detailed and clear requirements for software development tools, programming languages, coding standards, graphic standards, database naming and design specifications, etc. Under the constraints of a unified software architecture and technical specifications, it can ensure that the future developed software subsystems are consistent in terms of technical route, development tools, programming language, code standards, data resources, graphic standards, and interface style. Finally, it realizes continuous expansion of business functions, lever-type reuse of functional modules, inheritance and sharing of output results, thereby effectively improving system development efficiency, and promoting the system to continue integration, operation, and iterative upgrades towards platformization.

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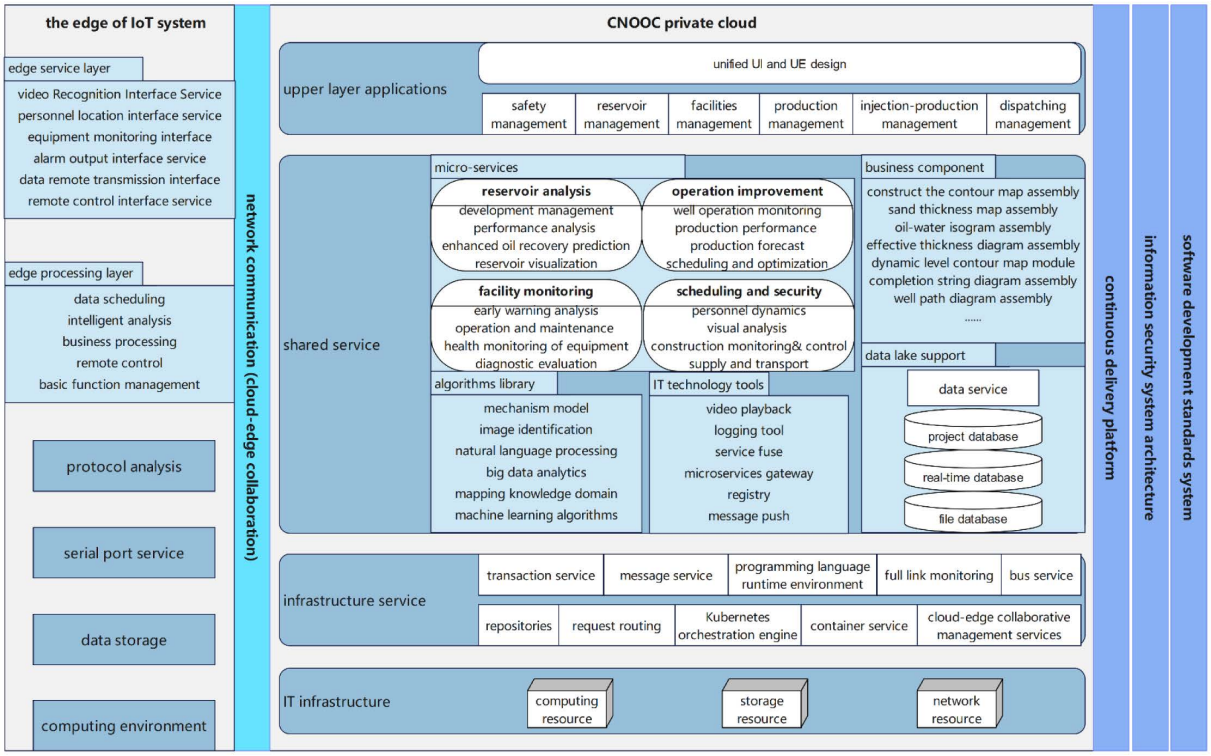

Figure 3—The overall architecture of offshore intelligent oilfield production monitoring system

The edge computing serves as the base, utilizing sensors and other devices to realize the perception and monitoring of all asset objects in the oilfield, and simultaneously collecting and transmitting relevant data of production and management to the back end for processing in real time. Algorithm modeling is the core. Based on the comprehensive modeling of objects such as reservoirs, oil and water wells, pipelines, key equipment and facilities, integrated simulation is conducted through data, topology, and other association relationships, laying a foundation for the optimization of various plans and measures. Visualization business components serve to display the static and dynamic data of various business domains in an intuitive form anytime and anywhere as needed, including the use of digital twin technology to make the production process and business collaboration transparent and visual, and support interactive processing.

Application architecture. The primary role of the production monitoring system is to maintain a vigilant watch over the performance of production-related machinery and processes in offshore oilfields. Its duty is to swiftly flag any anomalies, thereby ensuring the secure and uninterrupted functionality of all production apparatus and operations. This is manifested through its business applications which span across production management, facility management, dispatch management, and safety management. Moreover, the assembly of a real-time data integration and service environment actualizes instantaneous data collection and transmission at the edge level and the amalgamation of multi-source data at the cloud level (see Figure 4).

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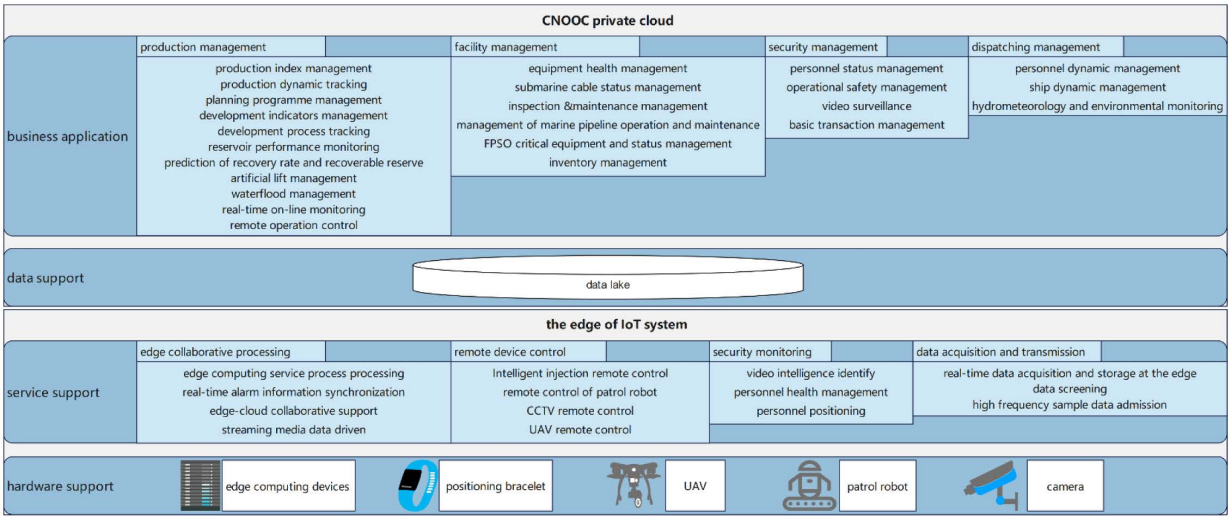

Figure 4—The application architecture of offshore intelligent oilfield production monitoring system

Technical architecture. Given that the system is mainly composed of the cloud services (IaaS, PaaS) and edge computing services, the technical architecture is divided into cloud based technical architecture and edge computing based technical architecture, combining the two through cloud-edge collaboration technology. The cloud based technical architecture is shown in Figure 5. The overall software development cloud management technology is divided into the following four parts:

1. Integration of underlying service resources, tenant permission control, and isolation of development, testing, and running software projects.
2. Centralized code repository management of project source code, realization of the complete process of source code storage, packaging, compilation, release, and warehouse management, ensuring the integrity and correctness of the code.
3. Project development and testing environment built on the PaaS private cloud management environment, providing isolated, elastic scalable development and testing environments according to application modules and computing services, customizing automated development, compilation, packaging, release, testing pipelines, achieving continuous integration and continuous delivery, unified and refined project management.
4. Service encapsulation through images; services can call related configuration files; eliminate the problems of environmental heterogeneity, achieve build once run anywhere, and can independently manage each step of application delivery.

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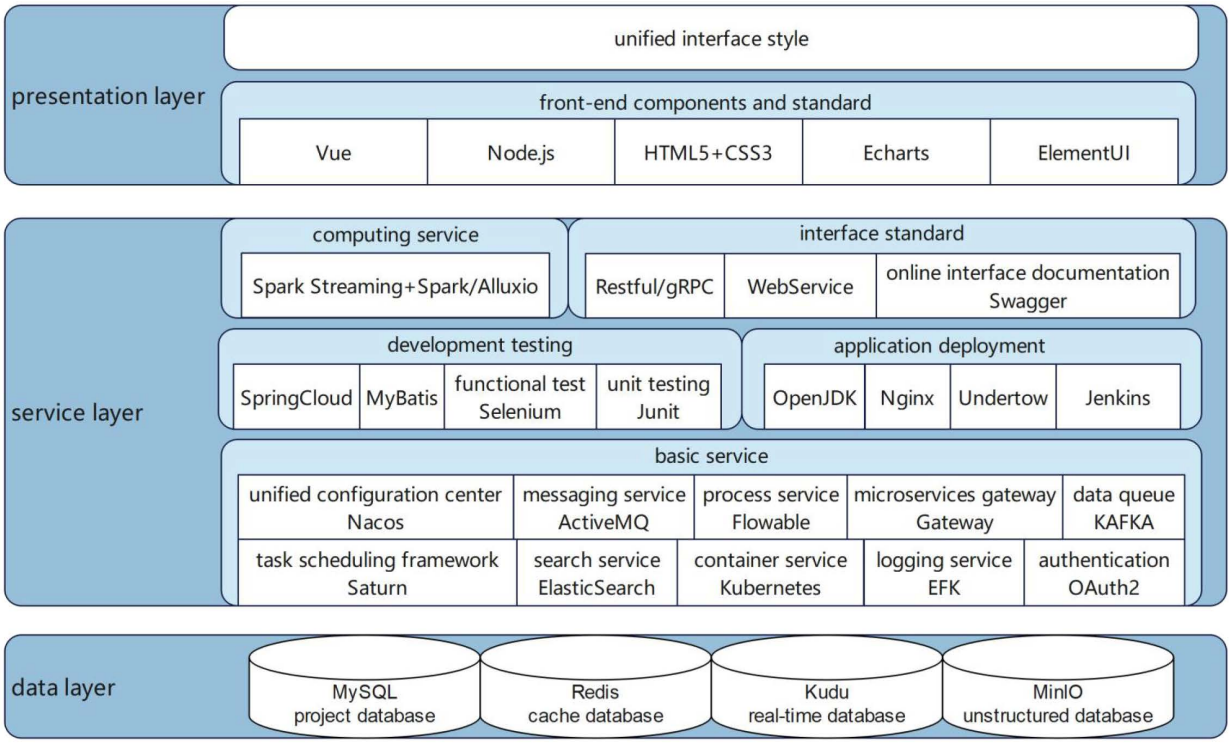

Figure 5—The cloud based technical architecture of offshore intelligent oilfield production monitoring system

The edge end establishes the hardware environment required for data docking, data screening, data remote transmission, and video recognition through the edge computing machine. It carries the real-time data access, transmission, and edge computing of offshore platforms, establishing a data return and edge end production monitoring recognition operating environment and computing capabilities. From a technical architecture perspective, the oil and gas production edge computing system consist of a perception layer, a processing layer, and a service layer (see Figure 6).

1. The perception layer consists of various perception devices and control devices installed on oil and gas production equipment, which, through transmitters, flow meters, and other sensors, automatically collects real-time data of oil, gas, and water wells, gathering and transporting pipelines and production monitoring, thereby realizing real-time perception of production status and automatic control of various equipment.
2. The processing layer mainly diverts the collected production data into edge computing nodes through cloud-edge intelligent collaboration technology, effectively reducing network load and data transmission delay. Meanwhile, the intelligent module built on the computing nodes enables intelligent screening and processing of production data. The cloud end is responsible for the training tasks of the computing analysis model and sends the trained models to the edge computing machine. The cloud and edge collaborate to solve the current problems of large delay in real-time data analysis and insufficient computing power, thereby improving user experience and realizing various localized artificial intelligence typical applications.
3. The service layer uses a combination of wired data transmission and wireless data transmission to transmit the production data, video, and images collected by on-site sensors and edge computing analysis data to the cloud.

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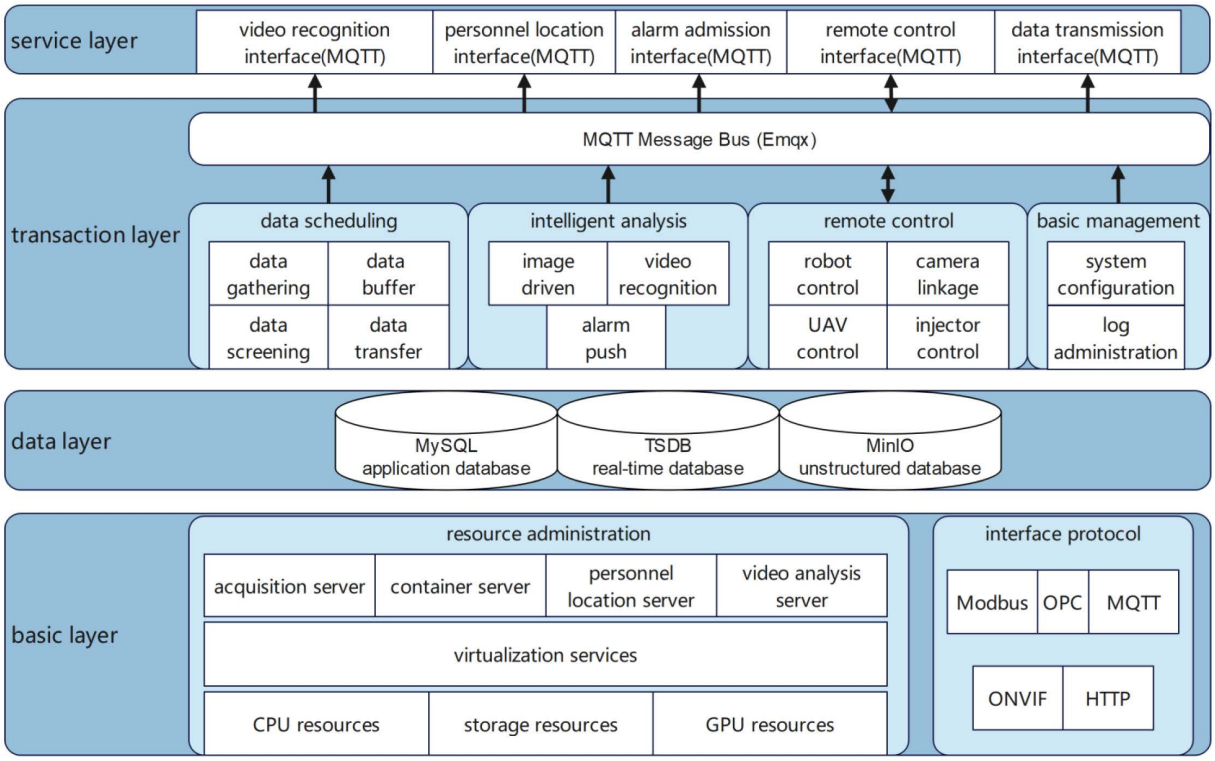

Figure 6—The edge computing based technical architecture of offshore intelligent oilfield production monitoring system

Data architecture. The data architecture consists of three parts: data source, data storage, and data services (Ref.5). The data architecture can realize unified management of data sources, centralized data storage and unified governance, while supporting intelligent data analysis and shared applications (see Figure 7).

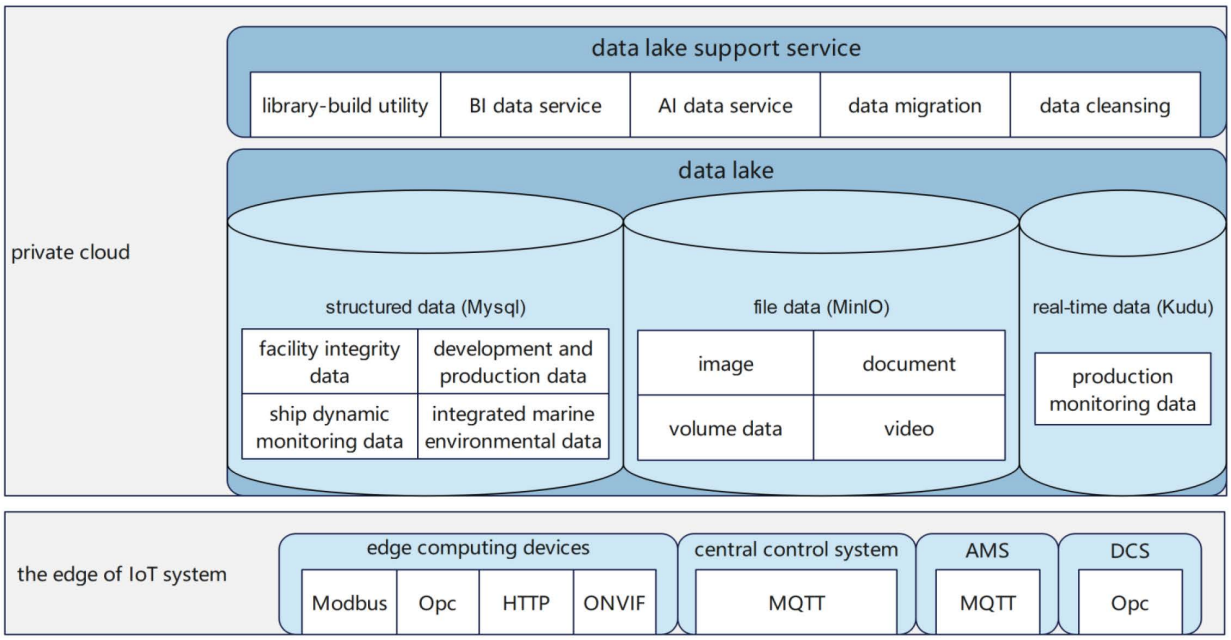

Figure 7—The data architecture of offshore intelligent oilfield production monitoring system

Integration architecture. During the system design process, it is necessary to integrate previously built and used heterogeneous systems with different functions, different structures, and different programming languages, such as the development production information system, equipment and facilities management

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system, electronic sea chart, vessel monitoring system, etc. For this reason, an integration architecture design was carried out (see Figure 8). To ensure good compatibility between systems, the following three principles of heterogeneous system integration need to be considered during the system development process.

1. Compatibility principle. The compatibility of interfaces, data, and architecture needs to be considered.
2. Communication principle. Communication protocols, network security, and communication performance need to be considered.
3. Data exchange principle: The data format, data accuracy, data security, and data processing performance need to be considered.

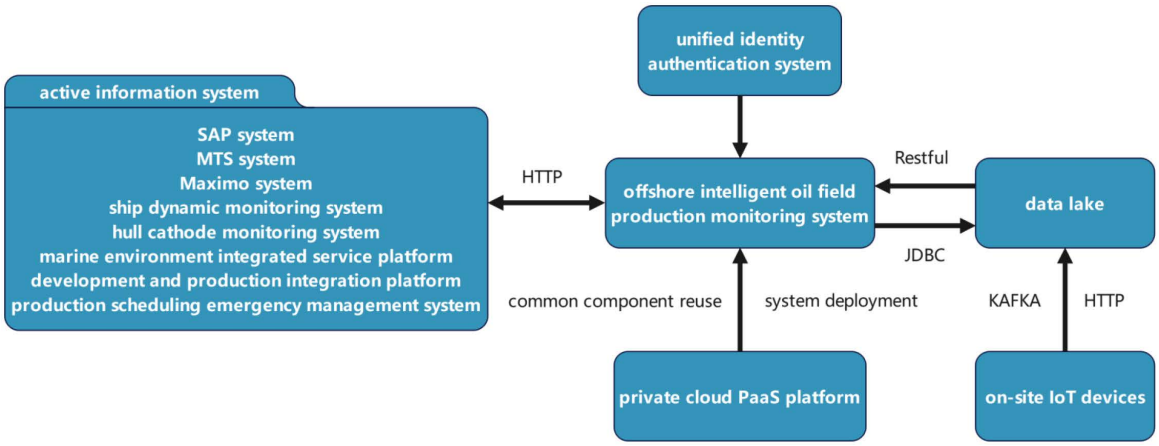

Figure 8—The integration architecture of offshore intelligent oilfield production monitoring system

Network security architecture. The technical defense measures that need to be reinforced in this system include security services oriented to applications and data, host protection oriented to the on-site, auditing and detection of the network transmission, regional isolation at the network boundary, centralized management of overall protection functions, and the supporting security management system construction (see Figure 9). A security management center is established to centrally control industrial control security products such as industrial firewalls, security audits, and host protection, centrally audit and analyze logs/alerts, turn daily security management work from disorder to order, simplify complexity, and comprehensively enhance network security management capabilities in the industrial control network and the production network. To ensure the safety of the production network, it is necessary to deploy boundary security protection equipment between the production network and the office network. The current approach is to deploy a net gap between the production network and the data collection server, which serves to physically isolate data and has a unidirectional function; A firewall is deployed between the data collection server and the office network to protect the TCP/IP protocol data transmission and block hacking attacks on the data collection server from the office network (Internet). To further reduce the security threats of the office network to the production network, formulate an overall security strategy according to the results of demand analysis:

1. Under the premise of ensuring business processes, achieve strict restricted access to the production network by office network users.
2. Strictly control the spread of malicious programs such as viruses and trojans to the production network by office network users.
3. Dynamic monitoring and real-time blocking of illegal and non-compliant network activities.
4. Strictly control the access rights of third-party vendors to the network management system.
5. Regularly train staff in information security to improve the information security awareness of relevant personnel.

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6. Regularly evaluate and review the overall system network to ensure network health.
7. Set different levels of protection strategies for each network domain.
8. Encrypt important information in the industrial control network and guarantee operation data security through digital certificate authentication.

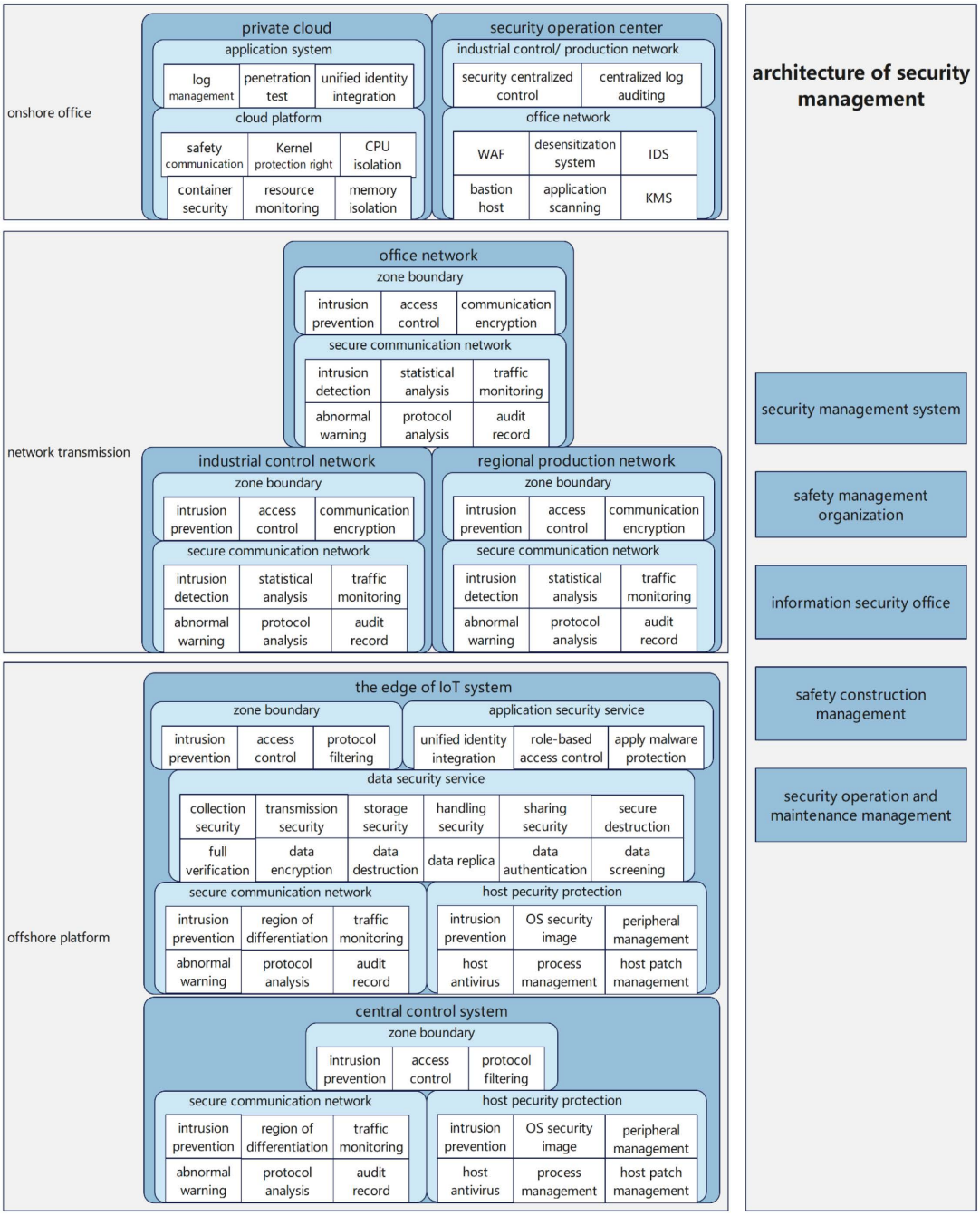

Figure 9—The network security architecture of offshore intelligent oilfield production monitoring system

Business function development and technical path

Oil and gas production management

The oil and gas production management subsystem relies on the technology of constructing an integrated "digital twin" of the reservoir, wellbore, and surface process that we have independently developed,

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combined with downhole parameters, oil and gas field production data, etc., to enhance the visualization monitoring capabilities of oil and gas production. We are developing risk monitoring technology for the process of oilfield development and production to achieve monitoring and prediction of risks in the development and production process, such as indicator changes, production fluctuations, reservoir management, injection and production allocation. With the oil and gas field development and production as the main line of business, focusing on reservoir operation management, developing functions such as dynamic reservoir management, dynamic injection and production management, and coordinated production process management, establishing a closed-loop management model of production operation "real-time collection-simulation-model diagnosis-expert decision-remote control", Improve the utilization degree of oil and gas field reserves, reduce the overall water cut, and slow down the decline in production (Ref.6).

Reservoir dynamic management. By integrating various types of geological data and dynamic monitoring data, applying professional geological reservoir research methods, and linking geological reservoir data, algorithm models, and analysis results, a comprehensive model that facilitates professional collaborative research is established. This model adheres to the business logic of well monitoring, single well/well group/block analysis, remaining oil distribution analysis, production index evaluation, production forecast, and measures recommendation. This aids in the potential excavation of oil reservoirs, adjustment optimization, and the enhancement of reservoir management levels (see Figure 10). The main functions are explained as follows:

1. Production management: By building oilfield production monitor models, it dynamically tracks the production change difference. This aids reservoir engineers by ensuring the effective achievement of annual distribution goals.
2. Well dynamic analysis: Based on the well dynamic analysis knowledge base and the knowledge graph technology, well dynamic analysis models are established. These models provide well production early monitoring and single well potential analysis.
3. Water injection dynamic analysis: Enables monitor for water injection rates and analysis of individual well injection potential. This term more accurately reflects the function of monitoring the performance of water injection, rather than production.
4. Well group dynamic analysis: Includes injection and production well group oil layer connection status analysis, well group injection-production balance analysis, and well group remaining oil distribution status analysis.
5. Oilfield dynamic analysis: Uses water drive characteristic curves for oilfield yield prediction.
6. Production monitoring: Provides monitor ability for production indices and production fluctuations during the production process, aids oil and gas production engineers in completing production analysis and production trend analysis.

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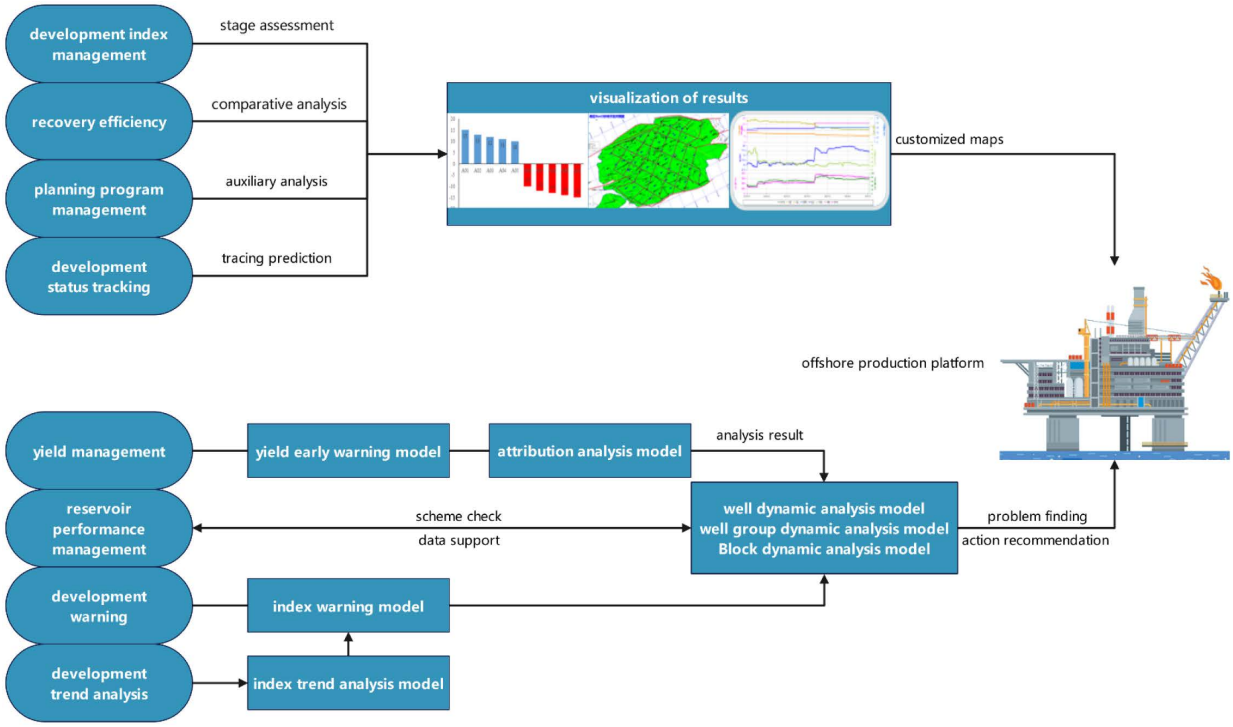

Figure 10—Reservoir management business logic description

Injection and production dynamic management. By constructing intelligent applications like artificial-lift well monitoring, electric pump failure prediction, and separated-zone water injection operation monitoring, it achieves running well group working condition monitoring, fault warning, auxiliary analysis, and work preplan. It uses data mining algorithms and artificial intelligence technology to analyze the direction and flow of injection and production fluid based on the calculation results of numerical simulation. It implements intelligent and automated development scheme push in accordance with balanced displacement. Based on the production situation of the well network, it conducts rapid injection and production adjustment optimization. This creates a management closed loop of injection and production process dynamic online monitoring, analysis, decision-making, and regulation. This offers scientific and intelligent methods and tools, and a reliable decision-making basis for further improving injection and production efficiency and oil and gas recovery (see Figure 11). The main functions are explained as follows:

1. Monitoring and warning: Based on decision trees and self-learning methods, a closed-loop management function for artificial-lift wells is achieved, which includes working condition monitoring, fault diagnosis, disposal measure recommendations. Based on big data analysis methods and mechanism analysis methods, a "one well, one model" health analysis diagnosis model is established for artificial-lift wells, realizing real-time operation monitoring, health status evaluation, abnormal situation monitoring, and auxiliary analysis.
2. Balanced deployment of surface processes: This evaluates whether the surface water injection system meets the distribution needs and provides early warning and optimization suggestions for water volume distribution.
3. Water well operation monitoring: Integrates the layered flow rate, pressure, and other real-time data of separated-zone water injection, tracks and analyzes the execution situation of layered geological distribution, implements layered water injection abnormal situation monitoring analysis, and information push.
4. Injection and production distribution: Uses comprehensive methods to split the yield of oil wells and water wells, calculates the inter-well connection coefficient model, realizes quantitative

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characterization of the inter-well connection of injection and production, and conducts planar fluid shunting.

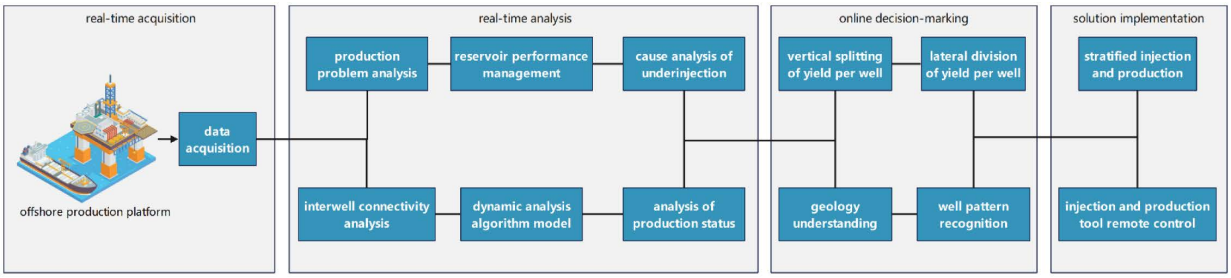

Figure 11—Injection-production management business logic description

Collaborative management of production processes. This section is about the use of core technology applications and the automatic correlation of data and models to realize dynamic perception, centralized display, and intelligent monitoring of production operations. This includes the establishment of an integrated dynamic analysis model from the oil well, the mixed transmission pipe network, the surface treatment, the water injection network to the water injection well and perfecting the online simulation application of surface production processes. The main functions are as follows:

1. Real-time production measurement: It sets up a multi-model fusion model for virtual measurement of each oil well, completes model algorithm testing and optimization, and performs automatic monitoring and analysis of the operating conditions of the entire oilfield oil well fluids, wellbore, ESP, chokes, and flowlines. This allows for low-cost, high-precision, real-time production monitoring and analysis of the entire oilfield, providing production data for other calculation modules.
2. Production schedule planning: By using virtual measurement off-line simulation, it simulates well pump change and other production conditions, calculates the corresponding production changes, and efficiently predicts the productivity of oil wells.
3. Dynamic operation of waterflooding recovery: It establishes a one-stop production process for oil well production, mixed transmission, platform processing, water injection network, and water injection pump. It allows for integrated real-time monitoring of the dynamic operation of waterflooding recovery in the oil field and comprehensive analysis of production, transmission, treatment, water injection, and strata absorption. It constructs an integrated allocation optimization method for oil and water logistics, conducts bottleneck judgment and feasibility analysis, and recommends optimization plans for increasing production and reducing consumption.

Equipment and facilities management

This section discusses integrating and invoking existing system data, designing professional algorithm models and functions in conjunction with equipment and facilities integrity management systems, accessing full-scale equipment real-time operation data, monitoring equipment and facilities status, predicting anomalies, and analyzing faults through edge computing, and cloud computing to improve the reliability of equipment and facilities.

Equipment health management. This section focuses on moving equipment, static equipment, electrical equipment, intelligent instruments, and central control equipment. We develop and apply various types of equipment health diagnosis model algorithms, realizes online monitoring of equipment operating status, intelligent early warning of faults, comprehensive diagnostic analysis, and proactive recommendation of maintenance strategies. This module, combined with inspection and maintenance management modules,

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helps to formulate scientific maintenance plans, optimize resource allocation, and improve the quality and efficiency of equipment operation and maintenance. The main functions include:

1. Moving equipment health management: It statistically analyzes and locates the status monitoring and fault diagnosis information of the water injection pump, and finally realizes fault statistics and early warning push with the water injection pump fault diagnosis model.
2. Static equipment management: It calls up relevant data from the offshore platform static equipment management system, analyzes the wall thickness thinning situation, cathodic protection potential information, etc., and graded early warns, alarms, and handles out-of-limit measurement points.
3. Electrical equipment management: It realizes real-time monitoring and analysis of electrical equipment such as transformers, medium and high voltage distribution cabinets, and low voltage distribution boards. It combines electrical equipment fault diagnosis algorithm models for multidimensional comprehensive evaluation.
4. Smart instrument management: It monitors the health status of smart instruments online and provides real-time predictive alarm information. It establishes a smart instrument health diagnosis model to comprehensively evaluate the health status of smart instruments.
5. Central control health management: It calls up the real-time diagnostic data of the central control system equipment, establishes a central control health diagnosis model for hardware diagnosis and software configuration management of the central control system, and realizes intelligent early warning and comprehensive diagnosis of the central control system.

Inspection and maintenance management. This section integrates static data from various systems about equipment and facilities to manage their inspection and maintenance records, including inspection plans, multi-dimensional queries for outsourced maintenance, performance indicator statistics, and maintenance record management. This aids in tracking the entire lifecycle of inspection and testing projects, assisting in the creation of maintenance plans, optimizing management resources, and providing effective support for predictive maintenance. Key functions are as follows:

1. Maintenance plan management: Statistically analyze the execution of inspection plans based on the type of equipment, inspection type, and planned completion time.
2. Performance assessment management: Aggregate equipment inspection and maintenance data, display the annual plan execution status, serviceability ratio, equipment failure impact, statistics of inspection and maintenance work order information, maintenance costs, etc.
3. Maintenance record management: Enable querying of detailed maintenance record information and historical information to enhance the efficiency of offshore user equipment maintenance work.
4. Inventory management: Implement statistical analysis features for key indicators such as account inventory, static inventory for three years or more, current year procurement material consumption rate, inventory aging, and in-transit inventory.

Submarine pipeline operation and maintenance management. This section involves the use of digital sea maps to display and dynamically simulate submarine pipeline information. It offers category-wise statistics, early warning tracing, and visualization interaction, simulates the operation status of pigging, and reasonably analyzes and optimizes submarine pipeline operation and maintenance solutions. Key functions are as follows:

1. Corrosion monitoring management: Analyze corrosion monitoring data (coupon, bypass, FSM), and give early warnings for exceeding corrosion rates. Display related corrosion monitoring information of submarine pipeline bypass monitoring, coupon monitoring, FSM monitoring, and analyze corrosion rate conditions.

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2. Inspection and maintenance management: Display operation records of internal inspection, pigging maintenance, and route survey information of submarine pipelines, and statistically analyze pigging, internal inspection, and external survey data.
3. Operation and maintenance plan management: Based on the corrosion monitoring and inspection maintenance module, implement the ultrasonic guided wave monitoring results, evaluate the analysis results and operation and maintenance plan, and link the monitoring statistics module.

Submarine cable status management. This section involves analyzing data such as the temperature, current carrying capacity, disturbance, and strain of submarine cables, introducing a comprehensive assessment algorithm for submarine cables, establishing a submarine cable assessment system, and assessing the health status of submarine cables. Key functions include:

1. Status monitoring management: Display in real time the cable core temperature, ambient temperature, and working current value of the cable. Call on submarine cable strain monitoring, disturbance monitoring, current carrying capacity monitoring, and burial depth survey data to monitor, aggregate, and analyze the operation status of the submarine cable.
2. Alarm information management: Analyze data such as submarine cable temperature, current carrying capacity, disturbance, strain, and burial depth from the access system, identify exceeding data, and give early warnings.
3. Comprehensive assessment management: Based on the results of submarine cable status monitoring, perform a comprehensive score for the submarine cable, judge the overall status of the submarine cable, predict the probability of in-service faults, and determine the in-service risk level.

FPSO operations management. This section focuses on the business needs of FPSO operation and production management, improves the safety management level of FPSO through the digital construction and upgrading of hulls, single-point mooring devices, oil and gas send-out condition. It establishes a comprehensive monitoring, warning, analysis, recommendation, and execution mechanism. The main features are as follows:

1. Hull motion monitoring: Using existing field monitoring equipment, it obtains FPSO hull posture and hull orientation information, implementing hull posture monitoring and bow direction monitoring.
2. Distance monitoring between single-point mooring device and FPSO hull: Based on GPS-RTK system, it monitors the distance between the bow and the single-point mooring device according to the position of single mooring device.
3. Mooring leg attitude and rotation angle monitoring: It shows the inclination data of the mooring leg inclinometer, implementing mooring leg posture monitoring; displays the angle data of the mooring leg angle meter, realizing the mooring leg rotation angle monitoring.
4. Oil and gas send-out distance and angle monitoring: It uses a 3D vision matching model algorithm to monitor and alert of the distance and angle between the FPSO and the shuttle tankers in real-time.
5. FPSO offloading process monitoring: It displays data such as offloading oil pumps and oil and offloading flowmeters under oil and gas send-out condition, realizing real-time dynamic monitoring and analysis of various process data under FPSO offloading.
6. Dynamic monitoring of FPSO offloading hose pressure: It displays the FPSO offloading hose pressure under oil and gas send-out condition, realizing real-time dynamic monitoring of FPSO offloading hose pressure.

Dispatch management

Based on existing systems like the offshore personnel registration system, CNOOC ship dynamic supervision system, and Bohai bay environment comprehensive service system, it realizes real-time

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dispatch information provision for offshore platforms and onshore control centers through functions such as personnel list statistical analysis, personnel dynamic tracking, maritime vessel position dynamics, real-time weather forecasting display. Main features are as follows:

1. Personnel dispatch management: Mainly for personnel statistics, personnel qualification management, job types and status, personnel shift plan, and personnel attendance management.
2. Real-time dynamic management of vessels: Track and count offshore vessels, including duty vessels and shift vessels serving offshore platforms, real-time display of position dynamics and navigation plans on sea charts (including today's plan, tomorrow's plan, departure plan, return plan), support ship position, query, trajectory query, and on-duty vessel scheduling function.
3. Meteorological and hydrological monitoring: Real-time display of hydrological, weather, and latest weather forecast information near the production platforms.

Operation safety management

This section uses edge computing devices, image recognition technology, and unmanned inspection equipment to achieve intelligent identification and alarm control of unsafe human behaviors, unsafe states of objects, and management vulnerabilities. Combined with electronic approval information of work permits, it achieves linkage application of video around the work point, risk identification; combined with the platform's 3D visualization technology, it intuitively displays the distribution and details of operations in various areas, realizes real-time viewing and process management of on-site operation construction personnel dynamics, work permits, etc., comprehensively displays and real-time statistics of operation information, and provides high-risk operation prompts.

Remote monitoring and inspection management. The remote monitoring and inspection management system uses technologies such as video surveillance and AI intelligent recognition to alert and locate hazards such as platform fires and unsafe factors such as not wearing safety helmets. It supports video alarm record query services, retrieves relevant alarm details, forms a closed-loop management mode of remote monitoring and collaborative operation, and realizes real-time monitoring warnings, abnormal alarms, and shutdowns. The main features are as follows:

1. Video surveillance: Through remote video surveillance, real-time display of live video images is achieved. Remote video inspections are carried out using video surveillance, real-time display of on-site monitoring images of various inspection areas on the platform, rapid reporting of abnormal information, inspection record traces, and assistance in on-site inspections.
2. Fire probe monitoring: The fire probe is linked to the video surveillance system. When an alarm occurs, the system immediately retrieves the corresponding video surveillance information for display, assisting relevant personnel to quickly understand the on-site situation.
3. Unmanned intelligent inspection: The use of unmanned intelligent inspection equipment realizes clock-in inspections, one-click upload of abnormal information, reminders for missed inspections, and ensures that inspection content is traceable, trajectories are followable, abnormalities are reportable, and overall enhances inspection control capabilities.

Personnel dynamic management. This section enables platform and individual data display, real-time display of platform personnel lists, personnel basic information, personnel certificate information, personnel location information, personnel vital sign information, effectively strengthening the refined management of offshore crew. The main features are as follows:

1. Personnel status overview: Real-time collection of personnel vital sign information is achieved through wearing intelligent monitoring devices equipped with intelligent monitoring bracelets, realizing health abnormality alarms.

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2. Personnel positioning: Combining personnel positioning with the platform's 3D visualization model, it realizes real-time dynamic tracking of platform employees, with multiple functions such as batch positioning, trajectory query, video linkage, one-click help, and fall alarm.

Operation construction management. This section connects key nodes of sea-land construction, realizes online closed-loop management of outsourced construction from land application to sea startup, planning, process supervision to completion evaluation, improves sea-land construction planning management efficiency and on-site operation safety control, and meets the full process standardization and standardization of construction. The main features are as follows:

1. Work permit management: Online application and approval of work permits, realizing online and standardized management of work permits.
2. Work process visualization: Utilizing personnel positioning and 3D visualization technology, it real-time displays construction video images and quickly grasps the platform construction operation situation.

Conclusion

Our research provides a detailed examination of the inaugural stage of an intelligent offshore oilfield pilot project undertaken by CNOOC in Bohai bay. This trailblazing endeavor is contributing towards an increased production and a reduction in operational and maintenance expenses by 10% for the oil and gas field. By fostering an environment conducive to data interoperability and application integration, the project is propelling the application of advanced technologies such as big data and artificial intelligence in offshore oil and gas operations. The success of this initiative is setting a precedent for the standardization and normalization of digital infrastructure within CNOOC, enhancing the enterprise's digital maturity and intelligent capabilities. Ultimately, these advancements are underpinning CNOOC's high-quality production, reinforcing its commitment to operational excellence.

In this article, the system architecture design is carried out for the current offshore oil and gas production business requirements and intelligent function requirements. It studies the management thinking of using the cloud-edge collaborative system to solve the sea-land linkage of offshore oil and gas field systems, proposes a new architecture design for the offshore intelligent oil field production monitoring system, supports the construction of CNOOC's future intelligent oilfield system through this architecture, ensures the robustness and reliability of the system, and can greatly reduce investment and construction costs.

For CNOOC, it is currently in the research and development exploration stage of intelligent oil field construction, and intelligence is mostly reflected in intelligent auxiliary management decision-making. In order to better play the role of informatization innovation technology in the digital transformation of intelligent oil field management, it is necessary to innovate thinking, use new technology, realize the organic integration of hardware technology and software technology, provide general software services, integrated services, operation and maintenance services, and cloud-edge management platform-level services based on technical capabilities, standards, and capital support, with the goal of deepening business applications, enhancing cloud-edge security capabilities, and creating a cloud service ecosystem.

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Nomenclature

IoT = Internet of Things
CNOOC = China National Offshore Oil Corporation
PCS = Process Control System

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DCS = Distributed Control System
SCADA = Supervisory Control and Data Acquisition
WHP = wellhead platform
EPP = Electric Power Platform
FPSO = Floating Production Storage and Offloading
OPC = OLE for Process Control
OLE = Object Linking and Embedding
PLC = Programmable Logic Controller
RTU = Remote Terminal Unit
TCP/IP = Transmission Control Protocol / Internet Protocol
C/S = Client/Server
DevOps = Development and Operations
IT = Information Technology
UI = User Interface
UE = Users Experience Design
UAV = Unmanned Aerial Vehicle
TSDB = Time Series Database
CCTV = Closed Circuit Television
IaaS = Infrastructure as a Service
PaaS = Platform as a Service
HTTP = Hypertext Transport Protocol
AMS = Asset Management Solutions
DCS = Distributed Control System
MQTT = Message Queuing Telemetry Transport
ONVIF = Open Network Video Interface Forum
BI = Business Intelligence
AI = Artificial Intelligence
MTS = Seagoing Personnel Dynamic Tracking Management System
JDBC = Java Database Connectivity
OS = Operating System
WAF = Web Application Firewall
IDS = Intrusion Detection System
KMS = Key Management Service
ESP = Electric Submersible Pump
FSM = Field Signature Method
GPS = Global Positioning System
RTK = Real Time Kinematic

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